

Section B: Define Study Time Periods

	Study Time Period	Label	Definition	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
0	STUDY_PD	Full study period	01 Jul 2015 - 30 June 2025 (inclusive)	Full study time period		OK19012026
1	ID_PD	Patient identification period	01 Jan 2016 - 30 June 2025 (inclusive)	Z		OK19012026
2						
3	INDEX	Index date	Date of the patient's earliest qualifying multiple myeloma diagnosis date	Patients are required to have at least 1 inpatient cancer visit for multiple myeloma (ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or at least 2 outpatient visits in any position on claim, on separate days within 90 days, during identification period. (Patients with 2 claims 30 or 60 days apart will also be flagged for consideration.) The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42 "MM dx code". Flags should also be made to distinguish which claims are non-diagnostic claims. A non-diagnostic claim refers to a claim for treatment of a condition (e.g., infusion) rather than diagnosis (see tab 49 "Diagnostic").		OK19012026
13						
14	Baseline	Baseline period	The baseline period is fixed and is the six months before the index date.	Patients must have six months of CE with medical or pharmacy benefits before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026
15	FU_PD	Follow up period 	The follow up period is fixed and is from the index date till the end of the patient identification period (inclusive).	Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled; consider medical and/or pharmacy benefits. Follow up period ends the earliest of death or the end of the study period (30 June 2025). Note: Patients with ≥3 months of CE with medical or pharmacy benefits from the index date (with no allowable gaps in enrollment) will be flagged for consideration for sensitivity or secondary analyses, and ending on the earliest of health plan disenrollment or the study period for patients who do not have an assigned date of death <3mos after the index date.		OK19012026
16						
17	e.g., PST24	e.g., 24-month post-index period	e.g., INDEX_DT through INDEX_DT +720			
18	etc.					
19						
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1.Cover

2.QC Review

3.Data Prep

4.Study Pop

5A.MMA_MED

5B.MAP_MED

6.LOT1_BASE

7.SCT

10.LOT1_BASE_END OK

11.LOT2_5_BASE

12.LOT2_5_BASE_END

13.SCT_CART_LOT_START

14.LOT2_5_Scenarios

40.CL

Section C: Source Data Preparation									
Instruction: Section C should ONLY be used to document decisions to the source data before the analytic file build. If decisions apply to a variable in the analytic file, document in the appropriate variable definition.									
#	Source Table/Variable	Situation Requiring Resolution	Action	Reasoning / Comments	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)			
1	Days to months conversion	Assuming 30.4375 day per month	3 months = 91 days, 6 months=183 days, 9 months=274 days, 12 months=365 days	Consistent with derivation other GSK Oncology studies				OK19012026	
2	ELIG_COH	Study team does not want to exclude smoldering MM patients. Optum built study cohort with the intention to exclude smoldering MM patients.	The team will additionally include patients who had ≤1 non-diagnostic medical claim for MM during the baseline period. These patients were excluded by Optum originally. If a patient has newly diagnosed or active MM, it would be likely that they would experience frequent activity, hence why we look for 2 non-diagnostic MM claims close together in the first step of the attrition table. A smoldering patient would have infrequent activity, hence why a claim for MM that falls outside of the 30, 60 or 90 days window requirement that's used to identify newly diagnosed patients is likely a smoldering patient.	Study team will consider excluding patients with a stand alone treatment code (rather than any non-diagnostic code) on an MM claim in the baseline period for future sensitivity analysis. This would not remove smoldering patients but rather ensure our patients were <u>newly</u> active MM patients.				OK19012026	
3	FU_PD	Original IE criteria from Optum only considers patients with at least 5 times of CE from index onwards.	Study team wants to keep criteria as broad as possible.	Will only require at least a day of CE post index, though will flag patients with 3 or more months of CE, to potentially subset on these patients for later consideration.				OK19012026	
4	MED_COND	Conditioning medications are included in MM-approved code list	Ignore these treatments for now.	The team is not particularly interested in conditioning therapy. These treatments potentially will be added back to the LOT at a later time.				OK19012026	
5	Identifying inpatient and non-inpatient claims	Approach 1: Use TOS_CD if conditions are not restricted on inpatient services associated with a hospitalization which requires an admission and discharge date. 1. Extract the records from T_Medical and T_CONFINEMENT datasets Inpatient → POS IN (21, 51, 61) or TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF_INPVIS, FAC_IP SNF) Outpatient → none of the above Approach 2: For anything related to services happening during a hospitalization with room and board, apply the condition on the CONF_ID because the majority of the work we do includes both diagnoses and medical services e.g. procedures, and when we talk about inpatient utilization, it is usually in the context of an admission and discharge. 1. Inpatient records should be restricted to cases where CONF_ID is not NULL, from the T_CONFINEMENT table, where this has associated admission and discharge dates. 2. The confinement table has summary information, so if users are pulling medical claims for specific inclusion, user should match on the CONF_ID variable, where it is not NULL in the confinement table. All other records without a CONF_ID or where CONF_ID is NULL should be considered non-inpatient.	Approach: We implement both Approach 1 and Approach 2 in combination (OR logic) per Optum CDM business rules guidance: Approach 1 (POSTOS) Inpatient if POS IN (21, 51, 61) OR TOS_CD IN (FAC_IP ACUTE, FAC_IP REHSNF, PROF_INPVIS, FAC_IP SNF) Approach 2 (CONF_ID) Inpatient if CONF_ID is NOT NULL and validated against T_CONFINEMENT table with both ADMIT_DATE and DISCH_DATE present Non-inpatient: All records not meeting either approach above A claim is classified as inpatient if either approach identifies it as inpatient. This maximizes capture of true inpatient encounters.	Rationale: Using both approaches together provides the most comprehensive inpatient identification. From our validation on the 2025q2 data Total patients (Step 0): 94951 Any inpatient event: 43439 (45.2%) Via POSTOS (Appr 1): 42618 (44.3%) Via CONF_ID (Appr 2): 40357 (42.0%) Both approaches: 39536 (41.1%) POSTOS only: 3082 (3.2%) CONF_ID only: 821 (0.9%) Outpatient only: 52757 (54.8%)				OK19012026	

